Cúram 8.2.1

Merative ™ Cúram Design System Guide 8.1.0

Note

Before using this information and the product it supports, read the information in Notices on page 53

© Merative US L.P. 2012, 2025

© Merative US L.P. 2012, 2025

Edition

This edition applies to Cúram 8.2.1.

© Merative US L.P. *2018, 2025*

© Merative US L.P. 2012, 2025

Contents

Note iii

Edition v

1 Cúram Design System 9

2 What's new and release notes for Cúram Design System 11

2.1 What's new in the Cúram Design System 11

2.2 Cúram Design System release notes 11

3 Prerequisites and supported software 13

4 Installing and getting started with the design system 19

5 Upgrading the design system 23

6 Developing your web application 25

6.1 Tutorial: Creating a page in your application 25

6.2 JavaScript™ development environment 31

6.3 Design system packages 31

6.4 Connecting to REST APIs 32

The RESTService utility 33

Connecting to REST APIs on Tomcat 36

Authenticating against REST APIs 37

7 Deploying your web application to a web server 39

7.1 Install and configure IBM® HTTP Server with WebSphere® Application Server 39

Generating an IBM® HTTP Server plug-in configuration 40

Configuring the IBM® HTTP Server plug-in 40

7.2 Install and configure Oracle HTTP Server with Oracle WebLogic Server 41

Installing Oracle HTTP Server and its components 42

Configuring the Oracle HTTP Server plug-in 43

7.3 Installing and configuring Apache HTTP Server 44

7.4 Building your web application for deployment 45

7.5 Deploying your web application to a web server 45

8 Troubleshooting and support 49

8.1 Citizen Engagement components and licensing 49

8.2 Citizen Engagement support strategy 50

8.3 Examining log files 50

Notices 53

Privacy policy 54

Trademarks 54

© Merative US L.P. 2012, 2025

© Merative US L.P. 2012, 2025

1 Cúram Design System

You can use the design system to develop your own custom web applications in addition to the standard Cúram web client. The design system provides the foundational packages for building accessible and responsive web applications. It consists of a React UI component library, React development resources, and a style guide for creating web applications.

The design system incorporates the US Web Design Standards and also supports additional CSS, utility classes, and a layout framework to enable teams to quickly build Section 508 compliant, responsive, and production-ready web applications.

Documentation versions

The online documentation applies only to the most recent version of the design system. To read the documentation in PDF format for earlier versions, see the Cúram PDF library.

© Merative US L.P. 2012, 2025

© Merative US L.P. 2012, 2025

2 What's new and release notes for Cúram Design System

Read about the what's new and release notes for the Cúram Design System .

2.1 What's new in the Cúram Design System

Read about the enhancements and improvements in the Merative™ Cúram Design System.

8.1.0 (19 March 2026)

No what's new updates in Cúram Design System for this release.

2.2 Cúram Design System release notes

Read about enhancements and bug fixes in the Cúram Design System.

For more information about compatibility with Cúram versions, see 3 Prerequisites and supported software on page 13.

To read older release notes select a previous Cúram documentation version.

8.1.0 (19 March 2026)

Combo box with large number of results extends into footer (SPM-137314)

Previously, in the IEG form `comboBox` component, a large number of search results caused the dropdown list to extend below the page and into the footer, causing layout overflow. This disrupted the page layout, hindered interaction with the dropdown, and obscured footer content.

Now, the `comboBox` dropdown dynamically repositions to prevent overflow below the footer. When results exceed viewport space, a scrollbar appears within the dropdown, allowing navigation of long lists without disrupting layout or obscuring footer content. (DT037096)

Month names are clipped in the date picker when using a text spacing override (SPM-138312)

Previously, when a visually impaired user enabled a text spacing override, some longer month names in the date picker were clipped. The date picker now supports WCAG text spacing requirements fully, and all month names display correctly without clipping when text spacing overrides apply. (DT037139)

Inline error text is perceived as regular text by High Contrast mode users (SPM-151203)

Previously, inline error messages were indistinguishable from regular text when using high contrast mode. Now, an error icon has been added to make such messages more distinguishable. (DT037464)

© Merative US L.P. 2012, 2025

Save and Exit button gets truncated when language is in French (SPM-151699)

Previously, the "Save and Exit" button was truncated when the screen was displayed in French at 400% zoom. This issue is now resolved and the “Save and Exit” button is no longer truncated. (DT037374)

© Merative US L.P. 2012, 2025

3 Prerequisites and supported software

Before you install or upgrade, review the prerequisites and supported software to ensure compatibility.

Cúram Platform

Cúram Platform is a prerequisite for developing and deploying your web application.

The Cúram Design System asset is released at more frequent intervals than Cúram and requires specific Cúram versions to benefit from the most recent server-side enhancements and bug fixes.

Note:

- From Cúram Design System 5.0.0 onwards, new features, server-side enhancements, and defect fixes are supported only in the most recent Cúram version lines. Security fixes and defect fixes are supported on Cúram 7.0.10-7.0.11.
- The Cúram Design System 3.x.x version line continues to be supported for security updates and critical defect fixes only on the older compatible version lines of Cúram, 7.0.10 -7.0.11.
- The Cúram Design System 2.6 version line continues to be supported for security updates and critical defect fixes only on the older compatible version lines of Cúram, 7.0.4 -7.0.9.

For more information about the support strategy, see 8.2 Citizen Engagement support strategy on page 50.

Note: From October 2019, new features, and server enhancements and defect fixes, are delivered only in the most recent version line.

Table 1: Compatibility with Cúram

A list of the asset versions and their compatible Cúram versions.

Asset versions	Compatible Cúram versions
8.1.0 8.0.0 7.11.2	• 8.2.1 for all new features, enhancements, and defect fixes. • 7.0.10-7.0.11 for security fixes and defect fixes.
7.11.1 7.11.0 7.10.1 7.10.0 7.9.0	• 8.2.0 for all new features, enhancements, and defect fixes. • 7.0.10-7.0.11 for security fixes and defect fixes.

© Merative US L.P. 2012, 2025

Asset versions	Compatible Cúram versions
7.8.0 7.7.1 7.7.0 7.6.2 7.6.1 7.6.0 7.5.0 7.4.0	• 8.1.3 for all new features, enhancements, and defect fixes. • 7.0.10-7.0.11 for security fixes and defect fixes.
7.3.1 7.3.0 7.2.0 7.1.0 7.0.2 7.0.1 7.0.0	• 8.1.2 for all new features, enhancements, and defect fixes. • 7.0.10-7.0.11 for security fixes and defect fixes.
6.3.1 6.3.0 6.2.3 6.2.2	• 8.1.1 for all new features, enhancements, and defect fixes. • 7.0.10-7.0.11 for security fixes and defect fixes.
6.2.1 6.2.0 6.1.4	• 8.1.0 for all new features, enhancements, and defect fixes. • 7.0.10-7.0.11 for security fixes and defect fixes.
6.1.3 6.1.2 6.1.1 6.1.0	• 8.0.3 for all new features, enhancements, and defect fixes. • 7.0.10-7.0.11 for security fixes and defect fixes.
6.0.2 6.0.1 6.0.0 5.3.2 5.3.1 5.3.0 5.2.2 5.2.1 5.2.0 5.1.0	• 8.0.2 for all new features, enhancements, and defect fixes. • 7.0.10-7.0.11 for security fixes and defect fixes.

© Merative US L.P. 2012, 2025

Asset versions	Compatible Cúram versions
5.0.0	• 8.0.1 for all new features, enhancements, and defect fixes. • 7.0.10-7.0.11 for security fixes and defect fixes.
4.1.4 4.1.3 4.1.2 4.1.1 4.1.0	• 8.0.1 for all new features, enhancements, and defect fixes.
4.0.3 4.0.2 4.0.1 4.0.0	• 8.0.0 for all new features, enhancements, and defect fixes.
3.0.10 3.0.9 3.0.8 3.0.7 3.0.6	• 7.0.11 iFix 5 for essential maintenance, security updates and critical defect fixes. • 7.0.10 iFix 8 for essential maintenance, security updates and critical defect fixes.
3.0.5 3.0.4	• 7.0.11 iFix 3 for all new features, enhancements, and defect fixes. • 7.0.10 iFix 7 for essential maintenance, security updates and critical defect fixes.
3.0.3	• 7.0.11 iFix 3 for all new features, enhancements, and defect fixes. • 7.0.10 iFix 6 for essential maintenance, security updates and critical defect fixes.
3.0.2	• 7.0.11 iFix 2 for all new features, enhancements, and defect fixes. • 7.0.10 iFix 5 for essential maintenance, security updates and critical defect fixes.
3.0.1 3.0.0	• 7.0.11 iFix 1 for all new features, enhancements, and defect fixes. • 7.0.10 iFix 4 essential maintenance, for security updates and critical defect fixes.
2.9.1 2.9.0	• 7.0.11 for all new features, enhancements, and defect fixes. • 7.0.10 iFix 3 for essential maintenance, security updates and critical defect fixes.

© Merative US L.P. 2012, 2025

Asset versions	Compatible Cúram versions
2.8.6 2.8.5 (Including the 2.8.4 internal release) 2.8.3 2.8.2 2.8.1 2.8.0 2.7.0	• 7.0.10 iFix 3 for all new features, enhancements, and defect fixes.

Node.js

Node.js is a prerequisite for installing the React application and for developing and deploying your web application.

Compatible Node.js versions.

Supported software	Version	Prerequisite minimum	Operating system restrictions
Node.js	22 LTS (latest) 20 LTS (latest)	20 LTS (latest)	No

Note: By default, Node 16 uses Node Package Manager (npm) 8 and Node 18 uses npm 9. To use either of these configurations, you must specify the npm option `legacy-peer-deps` in your project. The way that npm treats peer dependencies changed. The Cúram Universal Access Responsive Web Application is using the `legacy-peer-deps` option as a temporary fix while we work to remove this constraint. For more information about `legacy-peer-deps`, see npm Docs. The following steps outline how to configure the `legacy-peer-deps` option:

1. Create a *`.npmrc`* file at the root of your project.
2. Add the `legacy-peer-deps=true` content to the file.

Application server, web server, and database

Deploying the web application requires a web server in the Cúram topology. The following application server, web server, and database combinations are supported for developing and deploying your custom application.

- IBM® WebSphere® Application Server, IBM® HTTP Server or Apache HTTP Server, and IBM® Db2®
- IBM® WebSphere® Application Server, IBM® HTTP Server or Apache HTTP Server, and Oracle Database

© Merative US L.P. 2012, 2025

- Oracle WebLogic Server, Oracle HTTP Server or Apache HTTP Server, and Oracle Database

For more information about installing an application server or database for Cúram, see the .

HTTP servers

These HTTP servers are supported for deployment.

Compatible HTTP server versions

Supported software	Version	Prerequisite minimum	Operating system restrictions
IBM® HTTP Server	9.0	9.0.0.5	No
	8.5.5	8.5.5.9	No
Oracle HTTP Server	12.2.1.3.0 and future fix packs	12.2.1.3.190808	No
Apache HTTP Server	2.4 and future patches	2.4	No

Web browsers

The following browsers are supported for developing and deploying your web application.

New versions of Chrome, Firefox, Edge, and Safari are released more frequently than Internet Explorer, and updates are installed automatically by default for these browsers. Design system releases are tested on the latest versions of the browsers that are available at the start of Merative ™'s development cycle;

Note: Only stable Chrome releases are tested.

If a browser is tested and no issues are found, Merative ™ certifies that version. The prerequisites advise the version that is certified at each new product release. If Merative ™ cannot certify a version, you might need to revert to a previous, fully certified version. While Merative ™ supports customers with newer versions of the browsers than the last certified version, customers must know that those versions of the browsers are not fully tested.

Supported software	Version	Prerequisite minimum	Operating system restrictions
Apple Safari	14 and future fix packs	13	No
Google Chrome	91 and future fix packs	80	No
Microsoft™ Edge	91 and future fix packs	44	No
Mozilla Firefox	89 and future fix packs	73	No

© Merative US L.P. 2012, 2025

Accessibility

This accessibility software is certified.

Supported software	Version	Prerequisite minimum	Operating system restrictions	Browser
Freedom Scientific JAWS screen reader	2023 and future fix packs	2023	No	Microsoft Edge
Apple VoiceOver	Any version and future fix packs	Any version	Any version	Safari

Note: The combination of Microsoft Edge and JAWS 2023 is the only certified screen reader and browser combination.

Previous versions

To see the prerequisites and supported software for previous versions, see the Cúram PDF library.

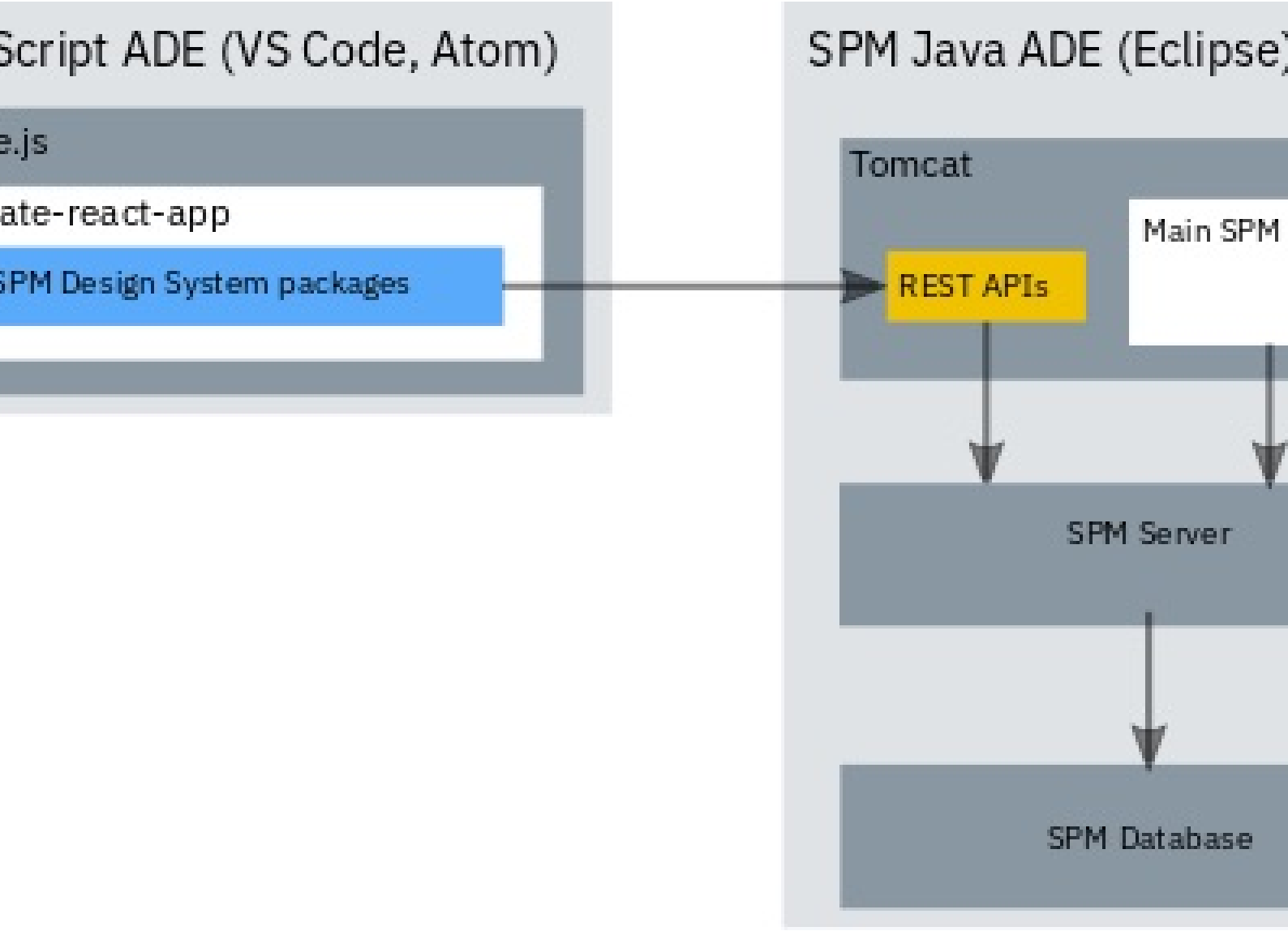
© Merative US L.P. 2012, 2025

4 Installing and getting started with the design system

To get started quickly, install Node.js, install a React application, install the Social Program Management Design System Node packages, and complete the short tutorial. When it is time to develop and test your REST APIs, install the Cúram Java Application Development Environment (ADE).

About this task

You need a React application into which you can install the Social Program Management Design System node packages. You can use any React application. However, the Facebook `create-react-app` contains some useful tools that simplify getting started with React development.



The Social Program Management Design System consists of the following Node packages:

- *`@spm/core`*

© Merative US L.P. 2012, 2025

- *@spm/core-ui*
- *@spm/core-ui-locales*
- *@spm/intelligent-evidence-gathering*
- *@spm-intelligent-evidence-gathering-locales*
- *@govhhs/govhhs-design-system-core*
- *@govhhs/govhhs-design-system-react*
- *@spm/eslint-config*
- *@spm/test-framework*
- *@spm/web-dev-accelerator-scripts*
- *@spm/web-dev-accelerator*

Attention: When working with npm packages, it is important that you familiarize yourself with the npm ecosystem and how package dependencies work, so that you can adopt a suitable security strategy for your project needs.

Procedure

1. Download and install Node.js from https://nodejs.org. The installation includes the npm (Node package manager), which you can use to install your Node packages.

   For more information about Node.js, see this Node.js developerWorks article.
2. Create your React application by using Facebook's *create-react-app*:
   a) From the directory where you want to create your React application, enter the following command, where *my-app* is the name that you want to call your application. A *my-app* directory is created to contain the application files.

   ```
   npx create-react-app my-app
   ```

   For more information about *create-react-app*, see the create-react-app user guide.

   b) You now have a basic React application in the *my-app* directory. You can run the application by entering the following commands:

   ```
   cd my-app
   npm start
   ```

   If the local host does not start automatically, browse to http://localhost:3000/ to see the running application.
3. Download the Cúram Design System Node packages. Open Cúram Support, under **Software Downloads**, select **Go to Downloads**, and follow the instructions to download the *SPM_DS_<version>.zip* archive file. Then extract the packages in the archive file to any directory.
4. Install and configure the Cúram Design System Node packages.
   a) From your *my-app* React application directory, enter the following command for each design system package to install the packages.

   ```
   npm install <path>/<package-name>-<version>.tgz
   ```

© Merative US L.P. 2012, 2025

Where *<path>* is the download path, *<package-name>* is a downloaded package, and *<version>* is the package version.

> **Note:** Ignore any Node package dependency warnings for now. If needed, you can resolve them later.

b) To import the minified JavaScript and CSS files for the design system components, edit the *my-app\src\App.js* file and insert the following lines:

```
import '@govhhs/govhhs-design-system-core/dist/js/@govhhs/govhhs-wds.min';
import '@govhhs/govhhs-design-system-core/dist/css/govhhs-wds.min.css';
```

5. Now, you can do the short tutorial to get started, or you can install the Cúram Java ADE that you need to develop and test your REST APIs, and do the tutorial later.

a) Complete the short tutorial to get started, see 6.1 Tutorial: Creating a page in your application on page 25.

b) Install the Cúram Java ADE, see Installing a development environmentthe .

© Merative US L.P. 2012, 2025

© Merative US L.P. 2012, 2025

5 Upgrading the design system

The design system is released on a frequent schedule. To upgrade to a new version, replace the Social Program Management Design System Node packages in your React application with the newer versions. Only the most recent version and two previous versions are supported. Ensure that you are reading the documentation for your version of the design system.

Procedure

1. Download the Cúram Design System Node packages. Open Cúram Support, under **Software Downloads**, select **Go to Downloads**, and follow the instructions to download the *SPM_DS_<version>.zip* archive file. Then extract the packages in the archive file to any directory.
2. Install the Cúram Design System Node packages. From your React application directory, enter the following command for each design system package to install the packages.

   In the command, *<path>* is the download path, *<package-name>* is a downloaded package, and *<version>* is the package version.

   > **Note:** Ignore any Node package dependency warnings for now. If needed, you can resolve them later.

   ```
   npm install <path>/<package-name>-<version>.tgz
   ```
3. Run the following command to install the package dependencies.

   ```
   npm install
   ```
4. Complete any upgrade steps in the 2.2 Cúram Design System release notes on page 11.

© Merative US L.P. 2012, 2025

© Merative US L.P. 2012, 2025

6 Developing your web application

Create your web application with the help of the development resources that are included in the Cúram Design System.

6.1 Tutorial: Creating a page in your application

In this short tutorial, you update the create-react-app *App.js* file in your React application to build an application page with the Social Program Management Design System components. For more information about how to design and use Social Program Management Design System components, see the Storybook documentation in *@govhhs/govhhs-design-system-react/doc/index.html*.

Procedure

1. Let's run the application. While it is running, you can see your changes as you make them. Run the application by entering the following commands:

```
cd my-app
npm start
```

   Browse to http://localhost:3000/ to see the running application.

2. Edit *my-app/src/App.js*.

First, let's add a header, which includes a nav bar, logo, and the agency name. For the header to load, we must import four components.

- Header
- Primary Navigation
- NavigationLink
- Link

3. We need to import the image for the header logo, so add the following line after the JavaScript and CSS imports lines we added during installation.

```
import logo from '@govhhs/govhhs-design-system-core/dist/img/logo-mark.svg';
```

   If a duplicate `import logo` statement is already included in the *App.js* file, delete it.

4. To make the components available for us to use, insert the following component import statement after the logo import statement.

```
import {
  Header,
  PrimaryNavigation,
  SecondaryNavigation,
  NavigationLink,
  Link,
  HeaderOverflowMenu,
  OverflowMenuOption
} from "@govhhs/govhhs-design-system-react"
```

© Merative US L.P. 2012, 2025

5. Replace the `<div>` code inside the create-react-app render function with the following code.

```
<div>
        <Header
          title="Agency name"
          logo={<img src={logo} alt="logo"/>}
        >
          <PrimaryNavigation>
            <NavigationLink
              title="Section One"
              id="basic-nav-section-one"
            >
              <Link>
              Sub-Link One
              </Link>
              <Link>
              Sub-Link Two
              </Link>
              <Link>
              Sub-Link Three
              </Link>
            </NavigationLink>
            <NavigationLink
              title="Section Two"
              id="basic-nav-section-two"
            >
              <Link>
              Sub-Link One
              </Link>
              <Link>
              Sub-Link Two
              </Link>
              <Link>
              Sub-Link Three
              </Link>
            </NavigationLink>
            <NavigationLink
              title="Link Three"
              id="basic-nav-section-three"
            />
            <NavigationLink
              title="Link Four"
              id="basic-nav-section-four"
            />
          </PrimaryNavigation>
        </Header>
       </div>
```

Add the Hero component.

6. In the component import statement, insert Hero into the list of imported components.

```
import {
  Hero,
  Header,
  PrimaryNavigation,
  SecondaryNavigation,
  SecondaryNavigationList,
  NavigationLink,
  Link,
  HeaderOverflowMenu,
  OverflowMenuOption
} from "@govhhs/govhhs-design-system-react"
```

7. When you add more than one component in the `return()` function, the JSX code must be wrapped in a surrounding `<div>` element. We want the Hero component to load underneath

© Merative US L.P. 2012, 2025

the header, so insert the `<Hero />` tag after the `</Header>` tag in the `return()` function. The `return()` function now has the following structure:

```
<div>
        <Header>
        ...
            <NavigationLink
              title="Link Four"
              id="basic-nav-section-four"
            />
          </PrimaryNavigation>
        </Header>
        <Hero />
     </div>
```

The hero component is displayed in the running application.

Now let's add some cards to our page. Cards are a great way to provide an entry point to more details, or access to actions on card content.

We must add some layout components to our list of imports:

- Grid
- Column
- Section

And add the card components to our list of imports:

- Card
- CardHeader
- CardBody
- CardFooter

8. Update your component import statement by inserting the following card components:

```
import {
  .....
  Card,
  CardHeader,
  CardBody,
  CardFooter,
  Grid,
  Column,
  Section,
} from "@govhhs/govhhs-design-system-react"
```

© Merative US L.P. 2012, 2025

9. To display the cards underneath the `</Hero>` component, append the following JSX code to lay three link cards out in a grid:

```
<Section>
         <Grid>
         <h2>Card Actions</h2>
         <p>Before using cards in your project, have a look at the guidance in
Storybook on how to use cards correctly.</p>
           <Column width="1/3">
             <Card href="/404" className="wds-u-mr--small wds-u-mt--small">
               <CardHeader title="Card header" highlight/>
                 <CardBody>
                   <p>
                     Neque porro quisquam est qui dolorem ipsum quia dolor sit
amet, consectetur, adipisci velit.
                   </p>
                 </CardBody>
               <CardFooter>
               </CardFooter>
             </Card>
             </Column>
             <Column width="1/3">
             <Card href="" className="wds-u-mt--small">
               <CardHeader title="Card header" highlight/>
                 <CardBody>
                   <p>
                     Neque porro quisquam est qui dolorem ipsum quia dolor sit
amet, consectetur, adipisci velit.
                   </p>
                 </CardBody>
               <CardFooter>
               </CardFooter>
             </Card>
             </Column>
             <Column width="1/3">
             <Card href="" className="wds-u-mt--small">
               <CardHeader title="Card header" highlight/>
                 <CardBody>
                   <p>
                     Neque porro quisquam est qui dolorem ipsum quia dolor sit
amet, consectetur, adipisci velit.
                   </p>
                 </CardBody>
               <CardFooter>
               </CardFooter>
             </Card>
           </Column>
         </Grid>
       </Section>
```

The cards are displayed in the running application.

Next, add a section that contains a form, an image, and a content area. For this section, we need to add some more components to our component import statement. Hopefully you're getting the hang of this!

- FieldSet
- TextInput
- Content
- TextArea
- Button
- Image

© Merative US L.P. 2012, 2025

10. Add the following items to your component import statement:

```
import {
  ....
  FieldSet,
  TextInput,
  Content,
  TextArea,
  Button,
  Image,
  Form
} from '@govhhs/govhhs-design-system-react';
```

11. Because we are using an image in this part of the page, we need to import that too. Add the following line before our component import statement:

```
import img from '@govhhs/govhhs-design-system-core/dist/img/hero-480.jpg';
```

12. For the last step, to append the JSX code to our `return()` function, add the following code after the last `</Section>` tag:

```
<Section className="wds-c-section--light">
          <Grid>
            <Column width="1/2">
            <Content>
              <h3 className="wds-u-mt--small">About</h3>
              <p>All design system components are fully responsive and will work
 on mobile, tablet and desktop browsers.
              Try it out by resizing the browser window.</p>
              <Image
                src={img}
                alt="Error loading Image"
                fallback={<span />}
              />
            </Content>
            </Column>
            <Column width="1/2">
              <h3 className="wds-u-mt--small">Form title</h3>
                <form>
                  <FieldSet>
                    <TextInput
                      label="Text input"
                      value="Input text"
                    />
                    <TextInput
                      label="Text input"
                      value="Input text"
                    />
                    <TextArea label="Text area label"></TextArea>
                    <Button category="primary" style={{float: "right"}}>Sign up</
Button>
                  </FieldSet>
                </form>
            </Column>
          </Grid>
        </Section>
```

Results

That's it, your page is now complete, you have created a fully responsive page in very little time! Hopefully you enjoyed this tutorial and see the benefits of using our design system.

© Merative US L.P. 2012, 2025

Example

Here's the final App.js file for reference:

```
import React, { Component } from 'react';
import './App.css';
import '@govhhs/govhhs-design-system-core/dist/js/@govhhs/govhhs-wds.min';
import '@govhhs/govhhs-design-system-core/dist/css/govhhs-wds.min.css';
import logo from '@govhhs/govhhs-design-system-core/dist/img/logo-mark.svg';
import img from '@govhhs/govhhs-design-system-core/dist/img/hero-480.jpg';
import {
  Hero,
  Header,
  PrimaryNavigation,
  SecondaryNavigation,
  SecondaryNavigationList,
  NavigationLink,
  Link,
  HeaderOverflowMenu,
  OverflowMenuOption,
  Card,
  CardHeader,
  CardBody,
  CardFooter,
  Grid,
  Column,
  Section,
  FieldSet,
  TextInput,
  Content,
  TextArea,
  Button,
  Image,
  Form
} from "@govhhs/govhhs-design-system-react";

class App extends Component {
  render() {
    return (
      <div>
        <Header
          title="Agency name"
          logo={<img src={logo} alt="logo"/>}
        >
          <PrimaryNavigation>
            <NavigationLink
              title="Section One"
              id="basic-nav-section-one"
            >
              <Link>
              Sub-Link One
              </Link>
              <Link>
              Sub-Link Two
              </Link>
              <Link>
              Sub-Link Three
              </Link>
            </NavigationLink>
            <NavigationLink
              title="Section Two"
              id="basic-nav-section-two"
            >
              <Link>
              Sub-Link One
              </Link>
              <Link>
              Sub-Link Two
              </Link>
              <Link>
              Sub-Link Three
              </Link>
            </NavigationLink>
            <NavigationLink
              title="Link Three"
              id="basic-nav-section-three"
            />
            <NavigationLink
              title="Link Four"
              id="basic-nav-section-four"
            />
          </PrimaryNavigation>
        </Header>
        <Hero />
        <Section>
          <Grid>
```

© Merative US L.P. 2012, 2025

6.2 JavaScript™ development environment

You can use any JavaScript™ development environment to develop your application, for example, Microsoft™ Visual Studio Code, Atom, or Sublime. Choose the tools that suits you best.

The Cúram Design System does not depend on any specific tools, so you can choose your own environment. However, Microsoft™ Visual Studio Code supports many plug ins that make development faster and easier, for example:

- Linting tools (ESLint)
- Code formatters (Prettier)
- Debugging tools (Debugger for Chrome)
- Documentation tools (JSDoc)

Merative ™ does not own, develop, or support any of the tools.

6.3 Design system packages

Use the design system packages to help you to develop and test your web client.

govhhs-design-system-core

This package contains a style guide, a library of user interface components, and front-end development resources that you can use to create Section 508-compliant, responsive, consistent web applications. The design system provides CSS, utility classes, and a grid framework so that you can quickly build accessible, responsive, production-ready websites.

govhhs-design-system-react

This package contains a React component library in Storybook to help you to build your application. It provides a collection of React components that align with the Cúram Design System.

For more information about the design guidelines, utility classes, and React components, see *`govhhs/govhhs-design-system-react/doc`*.

core

This package provides JavaScript™ utilities to help you develop your application. For example, use the *RESTService* utility to connect to a Cúram server-side REST API. Use `IntlUtils` to format numbers and dates for globalization.

For more information about the *`core`* package utilities, see the JSDoc API documentation in *`spm/core/doc`*.

core-ui

This package provides common React UI components to help you develop your application. For example, use the AppSpinner component to display a spinning animation while a page loads, or use the Toaster component to display notifications to the user.

© Merative US L.P. 2012, 2025

For more information about the `core-ui` components, see the JSDoc API documentation in `spm/core-ui/doc`.

core-ui-locales

This package provides translated artifacts for the `core-ui` components.

intelligent-evidence-gathering

This package enables IEG scripts that are configured in the Cúram application to run in your application. An API is provided to call the IEG scripts.

For more information, see the API documentation in `spm/intelligent-evidence-gathering/doc`.

intelligent-evidence-gathering-locales

This package contains translated artifacts for the `intelligent-evidence-gathering` package.

spm-web-dev-accelerator

This package contains the Web Development Accelerator rapid feature development tool, which generates Redux modules to handle the communication between your application and Cúram REST APIs.

spm-web-dev-accelerator-scripts

This package contains a Swagger parser to retrieve information from Cúram REST APIs, and scripts to generate the features and modules code from configuration information in the `spm-web-dev-accelerator` package.

spm-test-framework

This package contains a number of reusable files to help you to set up a test environment for testing with Test Cafe, Jest, and Enzyme. You can use the provided helper files to help you to develop and write end-to-end tests, unit tests, or snapshot tests for your project.

spm-eslint-config

This package contains an ESLint configuration with predefined coding style rules and an EditorConfig configuration file.

6.4 Connecting to REST APIs

You can connect your web application to REST APIs, such as the Cúram REST APIs.

Related information

© Merative US L.P. 2012, 2025

The RESTService utility

The *@spm/core* package provides the `RESTService` utility, which you can use to connect your application to a REST API. The `RESTService` utility provides important functions for securing and connecting to Cúram REST APIs, such as CSRF protection and SSO support. You can fetch resources with alternatives such as Fetch API, SuperAgent, or Axis, but you must consider implementing functionality that is handled by the `RESTService` utility, like CSRF protection and SSO support.

The `RESTService` utility supports the GET, POST, and DELETE HTTP methods through the following JavaScript methods:

- `RESTService.get(url, callback, params)`
- `RESTService.post(url, data, callback)`
- `RESTService.del(url, callback)`

See the full `RESTService` class documentation in the *doc* folder in the *@spm/core* package.

The `RESTService` utility hides details of calls, such as passing credentials, language, and errors. The callback that is passed to the GET, POST, or DELETE methods is started after the API calls return. API calls are asynchronous, so write your code to expect and handle a delay in receiving a response.

The `RESTService` utility provides functions during communications for authentication, handling responses, and user language.

Authentication

Authentication of the user is handled transparently by the `RESTService` utility. After a user is authenticated, the REST APIs automatically send the needed 'credentials', that is, the authentication cookies, with each request. For information about how authentication is handled for REST, see Social Program REST API securitythe .

If a user's session is invalidated before a new request is made to a REST API, then the '401 unauthorized' response is returned by the server. The `RESTService` utility relays the response to the callback function passed by the caller.

Handling responses

The `RESTService` utility formats the response from the server to ensure that callbacks receive the response in a consistent manner.

Each GET, POST, and DELETE method accepts a callback function from the caller. When called by the `RESTService` utility, the callback function receives a Boolean value that indicates the success or failure of the API call and the response. The callback function can then deal with the result. For example, a failure can be used to trigger your code to throw an error with the response data that can be used to trigger an error boundary. For more information about the callback function parameters, see the API documentation for the `RESTService` utility.

User Language

The 'Accept-Language' HTTP header is automatically set by the `RESTService` utility based on the user's selected language, which the user can select with the language picker in the application.

© Merative US L.P. 2012, 2025

This approach lets the server respond in the correct locale where locale sensitive information is being handled on the server.

The locale that is passed in the header is set in the transaction that is initiated by that REST request, and is used for the duration of that transaction. For more on transactions, see Transaction controlthe .

Cross-Site Request Forgery (CSRF)

The `RESTService` utility manages REST API CSRF protection for Universal Access that includes:

- Managing conditions on when to fetch a CSRF token.
- Pausing requests to fetch the CSRF token from the SPM server when needed.
- Storing the CSRF token in the application.
- Appending the CSRF token to the HTTP request header when appropriate.

Handling timeouts

The `RESTService` utility can manage unresponsive calls to the server. You can set environment variables in the *`.env`* files to set thresholds for timeouts.

- `REACT_APP_RESPONSE_TIMEOUT=10` Wait 10 seconds for the server to start sending.
- `REACT_APP_RESPONSE_DEADLINE=60` but allow 1 minute for the file to finish loading.

Simulating slow responses

During development, it is important to test that your application continues to operate in an acceptable way even when network responses are slow. You can simulate a slow network connection by setting a property in the *`.env.development`* file in the root of your project.

For example, set `REACT_APP_DELAY_REST_API=2` to delay the response from all GET requests for 2 seconds. The value can be set to any positive integer to adjust the delay.

Communicating with multiple API servers

Cúram Design System is configured to connect and communicate with a single Cúram server. However, it is possible to communicate with multiple API servers through the `RestService` utility.

About this task

To define where resource requests are directed, you can register a mapping function that is started by the `RESTService` utility before it makes the eventual call to the server. You can use the mapping function to map the requested resource, for example, *`/submitted_applications`*, to a specific endpoint, for example, `https://spm-server1.com`. For all other resources, the default server that is specified in the `REACT_APP_REST_URL` property is called. How the mapping is achieved depends on the mapping function, as shown in the following example.

© Merative US L.P. 2012, 2025

Note: Complexities with authentication and session management exist that you must resolve separately from configuring the server that requests are directed to. Resolving the authentication and session management issues are project-specific tasks, and are outside the scope of this documentation.

Procedure

1. Edit the *.env* configuration file in the root of your application, and add the REACT_APP_REST_URL1 and REACT_APP_REST_URL2 environment variables with the hostname and port of the server where the REST services are deployed, for example:

```
REACT_APP_REST_URL=/Rest
REACT_APP_REST_URL1=https://spm-server1.com/Rest
REACT_APP_REST_URL2=https://spm-server2.com/Rest
```

2. Update the *App.js* file to enable users to register and hook in a custom `apiEndPoints` function to determine the API endpoint URL for a particular resource, as shown in the following example:

```
import get from 'lodash.get';
   import REACT_APP_REST_URL from './REACT_APP_REST_URL';

   function isResourceMatchingDomainList(resource, domainItem) {
     const domainsDetailsList = JSON.parse(JSON.stringify(domainItem));
     const resArr = domainsDetailsList.map(x => x[Object.keys(x)[0]]);
     return resArr.find(y => resource.match(y));
   }

   const apiEndPoints = resource => {
     if (process.env.REACT_APP_REST_URL1 &&
       get(REACT_APP_REST_URL, 'data[0].REACT_APP_REST_URL1') &&
       isResourceMatchingDomainList(resource,
 REACT_APP_REST_URL.data[0].REACT_APP_REST_URL1) ) {
       return `${process.env.REACT_APP_REST_URL1}/${resource}`;
     }
     if (process.env.REACT_APP_REST_URL2 &&
       get(REACT_APP_REST_URL, 'data[0].REACT_APP_REST_URL2') &&
       isResourceMatchingDomainList(resource,
 REACT_APP_REST_URL.data[0].REACT_APP_REST_URL2) ) {
       return `${process.env.REACT_APP_REST_URL2}/${resource}`;
     }
     return `${process.env.REACT_APP_REST_URL}/${resource}`;
   };

   RESTService.registerApiEndPointsFunction(apiEndPoints);
```

© Merative US L.P. 2012, 2025

3. Create the *SPM-WebApps\packages\universal-access-sample-app\src\REACT_APP_REST_URL.js* file that contains the resource URL per domain to be retrieved, as shown in the following example:

```
const REACT_APP_REST_URL = {
     data: [
      {
         REACT_APP_REST_URL1: [
           {
             url: '^v1/activities',
           },
          {
            url: '^v1/ua/messages$',
          },
         {
           url: '^v1/ua/public_messages$',
          },
          {
          url: 'v1/ua/communications/',
         },
       ],
        REACT_APP_REST_URL2: [
          {
           url: '^v1/ua/appeals',
          },
        ],
      },
     ],
   };
   export default REACT_APP_REST_URL;
```

4. Use regex expressions to match the URL to the JSON defined content.
5. Enter the following command to install dependent packages:

```
npm install
```

6. Enter the following command to build the application into a build folder in the *universal-access-starter-pack*:

```
npm run build
```

7. Copy the build folder to the HTTP server and deploy it.

   For more information, see 7 Deploying your web application to a web server on page 39.

Connecting to REST APIs on Tomcat

If you have deployed REST APIs to Tomcat in your Eclipse development environment, you can connect to them from your application.

Before you begin

For more information about building and deploying Cúram REST APIs, see the *Cúram™ REST API Guide*.

© Merative US L.P. 2012, 2025

About this task

Use the API for Tomcat in the URL for your call to the RESTService utility. See the following example RESTService call.

```
RESTService.get('http://localhost:9080/Rest/v1/myAPI', (success, response) => {
  if (success) {
    // deal with response containing the json body
  } else {
    // deal with error contained in response
  }
});
```

Related information

Authenticating against REST APIs

You can authenticate against REST APIs by using the RESTService utility or by using the Cúram REST API security feature.

Authenticating against REST APIs by using the RESTService utility

Start the authentication URL *j_security_check* by using the POST function on the RESTService utility.

The following example shows a sample log-in invocation:

```
const callbackAfterLogin = (success, response) => {
  if (success) {
    // Login succeeded
  } else {
    // Login failed
  }
};
const loginUrl = 'http://localhost:9080/Rest/j_security_check';
const loginData = {
  j_username: username,
  j_password: password,
  user_type: 'EXTERNAL',
};
RESTService.post(loginUrl, loginData, callbackAfterLogin, 'form');
```

Authenticating against REST APIs by using the Cúram REST API security feature

Related information

Authenticating against REST APIs in Tomcat

A JAAS authentication mechanism is not exposed by the Tomcat and Eclipse environment, so you cannot authenticate your web application through the RESTService utility. To authenticate against a REST API in a Tomcat and Eclipse environment, use the Eclipse RMILoginClient class.

When you authenticate, log in by using your external user name and password to authenticate against the server. Subsequent calls to your REST API simulate this user.

Related information

© Merative US L.P. 2012, 2025

© Merative US L.P. 2012, 2025

7 Deploying your web application to a web server

You can deploy your web application on a web server in a production-like environment as part of your development process. Deployment in a production environment is outside the scope of this documentation, but you can use the instructions in this section for guidance.

7.1 Install and configure IBM® HTTP Server with WebSphere® Application Server

Install and configure IBM® HTTP Server either on the same server as WebSphere® Application Server or on a remote server. To enable cross-origin resource sharing (CORS), you can set the curam.rest.allowedOrigins property for the REST application on your application server, or install the IBM® HTTP Server plug-in for WebSphere® Application Server.

Before you begin

WebSphere® Application Server must be installed and configured.

You must install IBM® Installation Manager. For more information, see the IBM® Installation Manager documentation. You can download IBM® Installation Manager from Installation Manager and Packaging Utility download documents.

Note:

When the React application and the Cúram server are deployed in different hosts that don't share the same top-level domain+1, and the web server where the React app is hosted doesn't run a proxy plug-in towards the Cúram application servers, you must change the Cross-Site Request Forgery (CSRF) and session cookies for cross-origin requests, from the default `Samesite=Lax` to `Samesite=None`.

An alternative solution is to deploy a gateway web server in front of Cúram to modify the cookie by using this directive:

```
Header edit Set-Cookie ^(.*)$ $1;SameSite=None;Secure
```

For Cúram clusters, place this directive in the web servers where Cúram applications are mapped.

About this task

To enable cross-origin resource sharing (CORS), choose one of the following options:

- Set the curam.rest.allowedOrigins property for the REST application that is deployed on the application server. For more information about setting the curam.rest.allowedOrigins property, see Cúram REST configuration propertiesthe .

© Merative US L.P. 2012, 2025

- Install and configure the IBM® HTTP Server plug-in for WebSphere® Application Server to enable IBM® HTTP Server to communicate with WebSphere® Application Server. WebSphere® Customization Toolbox is needed to configure the plug-in.

Procedure

1. Install IBM® HTTP Server. For more information, see Migrating and installing IBM HTTP Server.
2. Optional: If you don't set the curam.rest.allowedOrigins property, you must install the following software:
   a) Install the IBM® HTTP Server plug-in for WebSphere® Application Server.
      For more information, see Installing and configuring web server plug-ins.
   b) Install the WebSphere® Customization Toolbox.
      For more information, see Installing and using the WebSphere Customization Toolbox.
3. Start IBM® HTTP Server. For more information, see Starting and stopping the IBM HTTP Server administration server.
4. To secure IBM® HTTP Server, see Securing IBM® HTTP Server.

Generating an IBM® HTTP Server plug-in configuration

This task is needed only if you install the IBM® HTTP Server plug-in for WebSphere® Application Server. Use WebSphere® Customization Toolbox to generate a plug-in configuration.

Before you begin

Start WebSphere® Application Server. For more information, see Starting a WebSphere® Application Server traditional server.

Procedure

To generate the IBM® HTTP Server plug-in configuration, complete the steps at the WebSphere® Application Server Network Deployment plug-ins configuration topic.

Configuring the IBM® HTTP Server plug-in

Configure the IBM® HTTP Server plug-in for WebSphere® Application Server and WebSphere® Customization Toolbox. This task is necessary only if you have chosen to install the IBM® HTTP Server plug-in, instead of setting the curam.rest.allowedOrigins property for the REST application that is deployed on the application server. Also, for information about how to configure the web server's HTTP verb permissions to mitigate verb tampering, see Enabling HTTP verb permissionsthe .

About this task

You can run the `configurewebserverplugin` target to complete the following tasks:

- Add the web server virtual hosts to the client hosts configuration in WebSphere® Application Server.

© Merative US L.P. 2012, 2025

- Propagate the plug-in key ring for the web server.
- Map the modules of any deployed applications to the web server.

Procedure

1. Start IBM® HTTP Server.

   For more information, see Starting and stopping the IBM® HTTP Server administration server.

2. On the remote WebSphere® Application Server, run the following command.

   ```
   build configurewebserverplugin -Dserver.name=server_name
   ```

   The `configurewebserverplugin` target requires a mandatory `server.name` argument that specifies the name of the server when the target is invoked. For more information about the `configurewebserverplugin` target, see Configuring a web server plug-in in WebSphere® Application Server.

3. Consider adding extra aliases to the *client_host*, as shown in the following examples:

   - For WebSphere® Application Server, add port number `9044`.
   - For the default HTTP port, add port number `80`.
   - For HTTPS ports, add port number `433`.

   For more information about client host setup, see step 19 in the WebSphere® Application Server port access setup topic.

4. To avoid port mapping issues from web applications, restart WebSphere® Application Server and IBM® HTTP Server.

   For more information, see Starting and stopping the IBM® HTTP Server administration server.

7.2 Install and configure Oracle HTTP Server with Oracle WebLogic Server

Install and configure Oracle HTTP Server on either the same server as Oracle WebLogic Server or on a remote server.

Before you begin

Oracle WebLogic Server must be installed and configured. For more information, see Installing and Configuring Oracle WebLogic Server and Coherence for Oracle HTTP Server 12.1.3, and Installing and Configuring Oracle HTTP Server for Oracle HTTP Server 12.2.1.3.

© Merative US L.P. 2012, 2025

Note:

When the React application and the Cúram server are deployed in different hosts that don't share the same top-level domain+1, and the web server where the React application is hosted doesn't run a proxy plug-in towards the Cúram application servers, you must change the Cross-Site Request Forgery (CSRF) and session cookies for cross-origin requests, from the default `Samesite=Lax` to `Samesite=None`.

An alternative solution is to deploy a gateway web server in front of Cúram to modify the cookie by using this directive:

```
Header edit Set-Cookie ^(.*)$ $1;SameSite=None;Secure
```

For Cúram clusters, place this directive in the web servers where Cúram applications are mapped.

Installing Oracle HTTP Server and its components

Install and configure Oracle HTTP Server in either a stand-alone domain, or in an Oracle WebLogic Server domain. You must install and configure an Oracle web server plug-in for proxying requests.

About this task

The Oracle web server plug-in allows requests to be proxied from Oracle HTTP Server to Oracle WebLogic Server. If you install and configure the Oracle web server plug-in, requests that are delegated to Oracle WebLogic Server still appear to originate from the Oracle HTTP Server, even if Oracle HTTP Server and Oracle WebLogic Server are hosted on two different servers.

Because of the web browser same-origin policy, cross-origin resource sharing (CORS) is restricted in many browsers by default. The web server plug-in enables CORS where Oracle HTTP Server and Oracle WebLogic Server are installed on different computers.

CORS enables an instance of your web application that is deployed on Oracle HTTP Server in one domain to request the REST services that are deployed on Oracle WebLogic Server in another domain.

Procedure

1. Install Oracle HTTP Server for Oracle WebLogic Server. For more information, see Installing and Configuring Oracle HTTP Server for Oracle HTTP Server 12.1.3, and Installing and Configuring Oracle HTTP Server for Oracle HTTP Server 12.2.1.3.
2. To configure Oracle HTTP Server, choose one of the following options:
   - To configure Oracle HTTP Server in a stand-alone domain, follow the instructions at Configuring Oracle HTTP Server in a Standalone Domain for Oracle HTTP Server 12.1.3, or Configuring Oracle HTTP Server in a Standalone Domain for Oracle HTTP Server 12.2.1.3.
   - To configure Oracle HTTP Server in an Oracle WebLogic Server domain, follow the instructions at Configuring Oracle HTTP Server in a WebLogic Server Domain for Oracle

© Merative US L.P. 2012, 2025

HTTP Server 12.1.3, or Configuring Oracle HTTP Server in a WebLogic Server Domain for Oracle HTTP Server 12.2.1.3.

3. If Oracle HTTP Server and Oracle WebLogic Server are installed in different domains, to enable CORS, install a web server plug-in.

   For more information about configuring an Oracle WebLogic Server proxy plug-in, see Configuring the Plug-In for Oracle HTTP Server for Oracle HTTP Server 12.1.3, or Configuring the Plug-In for Oracle HTTP Server for Oracle HTTP Server 12.2.1.3.

4. To secure Oracle HTTP Server, follow the procedure at Managing Application Security 12.1.3, or Managing Application Security for Oracle HTTP Server 12.2.1.3.

Results

The Oracle HTTP Server instance is now ready to for you to deploy the application. The default location for deploying the application is *`OHS_INSTANCE/config/fmwconfig/components/${COMPONENT_TYPE}/instances/${COMPONENT_NAME}/htdocs`*. However, you can configure the default location value to a different location.

What to do next

Start Oracle HTTP Server. For more information, see Next Steps After Configuring an Oracle HTTP Server Domain for Oracle HTTP Server 12.1.3, and Next Steps After Configuring the Domain for Oracle HTTP Server 12.2.1.3.

Configuring the Oracle HTTP Server plug-in

If a web server such as Oracle HTTP Server is configured in the topology, you must configure a web server plug-in in Oracle WebLogic Server. The web server plug-in enables Oracle WebLogic Server to communicate with Oracle HTTP Server. Also, for information about how to configure the web server's HTTP verb permissions to mitigate verb tampering, see Enabling HTTP verb permissionsthe .

About this task

To enable an Oracle HTTP Server web server plug-in in Oracle WebLogic Server, you can run the `configurewebserverplugin` target.

Procedure

1. Ensure the remote Oracle WebLogic Server Oracle WebLogic Server is running.

   For more information, see Deploying the application (Oracle WebLogic Server).

2. On the remote Oracle WebLogic Server, run the following command.

   The `configurewebserverplugin` target requires a mandatory `server.name` argument that specifies the name of the server when the target is invoked.

   ```
   build configurewebserverplugin -Dserver.name=server_name
   ```

   For more information about the `configurewebserverplugin` target, see Deploying the application (Oracle WebLogic Server).

3. Restart the remote Oracle WebLogic Server.

   For more information, see Deploying the application (Oracle WebLogic Server).

© Merative US L.P. 2012, 2025

7.3 Installing and configuring Apache HTTP Server

Install and configure Apache HTTP Server on either the same server as the application server or on a remote server. To enable cross-origin resource sharing (CORS), you can set the curam.rest.allowedOrigins property for the REST application on your application server, or install the appropriate plug-in for your web server. Also, for information about how to configure the web server's HTTP verb permissions to mitigate verb tampering, see Enabling HTTP verb permissionsthe .

Before you begin

An application server must be installed and configured.

Note:

When the React application and the Cúram server are deployed in different hosts that don't share the same top-level domain+1, and the web server where the React application is hosted doesn't run a proxy plug-in towards the Cúram application servers, you must change the Cross-Site Request Forgery (CSRF) and session cookies for cross-origin requests, from the default `Samesite=Lax` to `Samesite=None`.

An alternative solution is to deploy a gateway web server in front of Cúram to modify the cookie by using this directive:

```
Header edit Set-Cookie ^(.*)$ $1;SameSite=None;Secure
```

For Cúram clusters, place this directive in the web servers where Cúram applications are mapped.

About this task

To enable cross-origin resource sharing (CORS), choose one of the following options:

- Set the curam.rest.allowedOrigins property for the REST application that is deployed on the application server. For more information about setting the curam.rest.allowedOrigins property, see REST configuration propertiesthe *Developing Outbound REST APIs Guide*.
- Install and configure the plug-in for your server.

Procedure

1. Install Apache HTTP Server. For more information, see Compiling and Installing in the Apache HTTP Server documentation.
2. Optional: If you don't set the curam.rest.allowedOrigins property, you must choose one of the following options:
   - WebSphere® Application Server

     Install the plug-in for WebSphere® Application Server, see Installing and configuring web server plug-ins.

© Merative US L.P. 2012, 2025

Install the WebSphere® Customization Toolbox, see Installing and using the WebSphere Customization Toolbox.

To configure Apache HTTP Server with WebSphere® Application Server, see Configuring Apache HTTP Server.

- Oracle WebLogic Server 12cR1 (12.1.3):

  For more information about configuring an Oracle WebLogic Server proxy plug-in, see Configuring the Plug-In for Oracle HTTP Server.

  To configure Apache HTTP Server with Oracle WebLogic Server, see Configuring the Plug-In for Apache HTTP Server.
- Oracle WebLogic Server 12cR2 (12.2.1.3):

  For more information about configuring an Oracle WebLogic Server proxy plug-in, see Configuring the Plug-In for Oracle HTTP Server.

  To configure Apache HTTP Server with Oracle WebLogic Server, see Configuring the Plug-In for Apache HTTP Server.

3. Start Apache HTTP Server. For more information, see Starting Apache in the Apache HTTP Server documentation.
4. To secure Apache HTTP Server server, see Security Tips and Apache SSL/TLS Encryption in the Apache HTTP Server documentation.

7.4 Building your web application for deployment

You must build your web application for deployment on a web server. A *`build`* directory is created that contains all of the required files for your web application.

About this task

For more information about npm build and deployment, see npm run build in the `create-react-app` GitHub documentation.

Procedure

From your application root directory, run the following command to create the *`build`* directory.

```
npm run build
```

7.5 Deploying your web application to a web server

To test your web application against an existing Cúram application that is deployed on an enterprise application server, you can deploy the web application on one of the supported web

© Merative US L.P. 2012, 2025

servers. The supported web servers are all based on Apache HTTP server so the deployment procedure is similar.

Before you begin

You must have built your application for deployment.

About this task

The `universal-access-starter-pack` package includes a preconfigured *`.htaccess`* file under the public folder that gets added to the built application. This file contains comments to explain the web server configuration requirements for React Router `BrowserRouter` enablement.

For more information about how to configure *`.htaccess`* files in a web server, see the *Apache HTTP Server Tutorial: .htaccess files* related link.

For more information about React Router `BrowserRouter`, see Serving Apps with Client-Side Routing.

Procedure

1. Copy the contents of the *`build`* directory to the appropriate directory for your HTTP server.

   For more information about the `<directory>` directive, see the related links.
2. Configure the web server.
   - If you use *`.htaccess`*, enable the directives in *`.htaccess`* by editing *`httpd.conf`* and setting an appropriate value for the `AllowOverride` directive in the Directory section for the HTTP server's `DocumentRoot`, or the corresponding directory where the resources are being deployed.

     In addition, you must load the `mod_rewrite` module for the React Router `BrowserRouter`.

     ```
     # Enables mod_rewrite for React Router's BrowserRouter directives
     <IfModule !mod_rewrite.c>
         LoadModule rewrite_module modules/mod_rewrite.so
     </IfModule>
     # "/opt/IBM/HTTPServer/htdocs/universal" is the location
     # where the web application is deployed under the DocumentRoot.
     # Alternatively you can specify the DocumentRoot "/opt/IBM/HTTPServer/htdocs"
     <Directory "/opt/IBM/HTTPServer/htdocs/universal">
         AllowOverride FileInfo Options=MultiViews
     </Directory>
     ```
   - If you do not use `.htaccess`, you can copy the directives in *`.htaccess`* and put them in a `LocationMatch` section for your application in *`httpd.conf`*.

     ```
     # Enables mod_rewrite for React Router's BrowserRouter directives
     <IfModule !mod_rewrite.c>
         LoadModule rewrite_module modules/mod_rewrite.so
     </IfModule>
     # Below LocationMatch is set to "/universal" because the application
     # will be served from https://youhostname.com/universal
     <LocationMatch /universal>
         #
         # place here your .htaccess directives
         #
     </LocationMatch>
     ```

© Merative US L.P. 2012, 2025

3. Tune your HTTP server for improved performance, see the Performance Tuning guide.

Related information

GitHub documentation: npm run build
Content Security Policy Quick Reference Guide
Apache core features V2.0: <Directory> Directive
Apache core features V2.4: <Directory> Directive
Apache HTTP Server Tutorial: .htaccess files

© Merative US L.P. 2012, 2025

© Merative US L.P. 2012, 2025

8 Troubleshooting and support

Use this information to help you to troubleshoot issues with the Merative ™ Cúram Universal Access Responsive Web Application or Cúram Design System.

The Cúram supported assets can be installed, customized, and deployed separately from Cúram, before being integrated into the system.

When troubleshooting web applications that are integrated with Cúram, use this troubleshooting information in conjunction with the troubleshooting information for Cúram.

Related information

8.1 Citizen Engagement components and licensing

You can use and customize the Merative ™ Cúram Universal Access Responsive Web Application for your organization, or develop your own custom web applications in addition to the standard Cúram application. Use this information to understand the Cúram components, supported assets, and licenses that you need.

Installable components

- **Cúram Design System supported asset**
  The design system provides foundational packages for building accessible and responsive web applications. It consists of a React UI component library, React development resources, and a style guide for creating web applications.
- **Merative ™ Cúram Universal Access Responsive Web Application supported asset**
  The Merative ™ Cúram Universal Access Responsive Web Application provides a reference web application, which you can use and customize for your organization. The Merative ™ Cúram Universal Access Responsive Web Application requires the Cúram Design System and the Universal Access application module.
- **Universal Access application module**
  The Universal Access (UA) application module provides the Universal Access administrator application and the Universal Access REST APIs that expose interfaces to Universal Access functions for consumption by the Merative ™ Cúram Universal Access Responsive Web Application. Universal Access requires the Cúram Platform.

Licensing Universal Access

You can buy the Universal Access application module, which entitles the Merative ™ Cúram Universal Access Responsive Web Application asset, and Cúram Platform, which entitles the Cúram Design System asset.

Alternatively, you can buy Citizen Engagement, which includes the Universal Access application module, the Cúram Platform, and both assets.

© Merative US L.P. 2012, 2025

Licensing the Cúram Design System

To develop custom web applications to complement the Cúram Platform, you can buy the Cúram Platform, which entitles the Cúram Design System asset.

8.2 Citizen Engagement support strategy

The Citizen Engagement assets are typically released monthly, and they can be upgraded independently of Cúram . Each release is a full release and not a delta release.

The assets are supported for the lifetime of the latest supported Cúram version available at the time of the asset release.

- The main asset line is released monthly and contains new features, enhancements, security updates, defects, and support for the latest Cúram version.
- Merative ™ Cúram Universal Access Responsive Web Application 2.6 continues to be supported with security updates and critical defect fixes for older compatible Cúram versions.

Although new features can be delivered in any asset release, they are typically delivered at the same time as the Universal Access application module release that contains the new APIs for those features. Where possible, Universal Access REST API changes are delivered in refresh pack or other impact-free releases that impose no forced upgrade impact.

Semantic versioning

The assets use semantic versioning. As a general guideline, this means:

- MAJOR version for incompatible API changes
- MINOR version for adding functionality in a backwards-compatible manner
- PATCH version for backwards-compatible bug fixes

8.3 Examining log files

Log files are a useful resource for troubleshooting problems.

Examining the browser console logs

For JavaScript applications, you can examine the browser console logs for errors that might be relevant to investigating problems. For the exact details about how to locate the console logs within the browser, see your browser documentation.

Note: When you are developing applications with the Cúram Design System, console logging information might also be displayed in the console that runs the start process for the application.

Examining the HTTP Server log files

When you deploy a built application on an HTTP Server, the built application introduces a new point with which logging is captured in your system topology. The IBM® HTTP Server, Oracle

© Merative US L.P. 2012, 2025

HTTP Server, and the Apache HTTP Server include comprehensive logging system and related information.

For more information about troubleshooting the IBM® HTTP Server, see Troubleshooting IBM HTTP Server.

For more information about troubleshooting the Oracle HTTP Server, see Managing Oracle HTTP Server Logs.

For more information about troubleshooting the Apache HTTP Server, see Log Files.

Examining the IEG log files

System administrators can enable improved logging by setting the curam.trace system administration property to `trace_on` or higher, and you can then check the server logs after you call the datastore prepopulation feature. You can view detailed logs that are generated during the population of data during screening, application intake, and life events to better explain what interactions have taken place. Information is output to the server logs during datastore prepopulation to describe which code path was taken and why.

The following information is written to the server logs during datastore prepopulation:

- Information about which code path was taken and why.
- The values of the relevant system administration properties.
- The schema names of the relevant IEG scripts.
- The number of records in the ViewProcessor table.

© Merative US L.P. 2012, 2025

© Merative US L.P. 2012, 2025

Notices

Permissions for the use of these publications are granted subject to the following terms and conditions.

Applicability

These terms and conditions are in addition to any terms of use for the Merative website.

Personal use

You may reproduce these publications for your personal, noncommercial use provided that all proprietary notices are preserved. You may not distribute, display or make derivative work of these publications, or any portion thereof, without the express consent of Merative

Commercial use

You may reproduce, distribute and display these publications solely within your enterprise provided that all proprietary notices are preserved. You may not make derivative works of these publications, or reproduce, distribute or display these publications or any portion thereof outside your enterprise, without the express consent of Merative.

Rights

Except as expressly granted in this permission, no other permissions, licenses or rights are granted, either express or implied, to the publications or any information, data, software or other intellectual property contained therein.

Merative reserves the right to withdraw the permissions granted herein whenever, in its discretion, the use of the publications is detrimental to its interest or, as determined by Merative, the above instructions are not being properly followed.

You may not download, export or re-export this information except in full compliance with all applicable laws and regulations, including all United States export laws and regulations.

MERATIVE MAKES NO GUARANTEE ABOUT THE CONTENT OF THESE PUBLICATIONS. THE PUBLICATIONS ARE PROVIDED "AS-IS" AND WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO IMPLIED WARRANTIES OF MERCHANTABILITY, NON-INFRINGEMENT, AND FITNESS FOR A PARTICULAR PURPOSE.

Merative or its licensors may have patents or pending patent applications covering subject matter described in this document. The furnishing of this documentation does not grant you any license to these patents.

Information concerning non-Merative products was obtained from the suppliers of those products, their published announcements or other publicly available sources. Merative has not tested those products and cannot confirm the accuracy of performance, compatibility or any other claims related to non-Merative products. Questions on the capabilities of non-Merative products should be addressed to the suppliers of those products.

Any references in this information to non-Merative websites are provided for convenience only and do not in any manner serve as an endorsement of those websites. The materials at those

© Merative US L.P. 2012, 2025

websites are not part of the materials for this Merative product and use of those websites is at your own risk.

This information contains examples of data and reports used in daily business operations. To illustrate them as completely as possible, the examples include the names of individuals, companies, brands, and products. All of these names are fictitious and any similarity to actual people or business enterprises is entirely coincidental.

The licensed program described in this document and all licensed material available for it are provided by Merative under terms of the Merative Client Agreement.

COPYRIGHT LICENSE:

This information contains sample application programs in source language, which illustrate programming techniques on various operating platforms. You may copy, modify, and distribute these sample programs in any form without payment to Merative, for the purposes of developing, using, marketing or distributing application programs conforming to the application programming interface for the operating platform for which the sample programs are written. These examples have not been thoroughly tested under all conditions. Merative, therefore, cannot guarantee or imply reliability, serviceability, or function of these programs. The sample programs are provided "AS IS", without warranty of any kind. Merative shall not be liable for any damages arising out of your use of the sample programs.

Privacy policy

The Merative privacy policy is available at https://www.merative.com/privacy.

Trademarks

Merative ™ and the Merative ™ logo are trademarks of Merative US L.P. in the United States and other countries.

IBM®, the IBM® logo, and ibm.com® are trademarks or registered trademarks of International Business Machines Corp., registered in many jurisdictions worldwide.

Adobe™, the Adobe™ logo, PostScript™, and the PostScript™ logo are either registered trademarks or trademarks of Adobe™ Systems Incorporated in the United States, and/or other countries.

Oracle and Java are registered trademarks of Oracle and/or its affiliates.

The registered trademark Linux® is used pursuant to a sublicense from the Linux Foundation, the exclusive licensee of Linus Torvalds, owner of the mark on a worldwide basis.

Microsoft™, Windows™, and the Windows™ logo are trademarks of Microsoft™ Corporation in the United States, other countries, or both.

UNIX™ is a registered trademark of The Open Group in the United States and other countries.

Other company, product, and service names may be trademarks or service marks of others.

© Merative US L.P. 2012, 2025